

RegComEvidenceMatch at the NTCIR-19 RegCom Task: Evidence-Aware Multilingual Verification for ESG Compliance

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Abstract

Sustainability standards require evidence that is not only topically relevant but also complete, category-compatible, and expressed in the required unit. Yet multilingual ESG reports are long and semi-structured, and compliance systems must connect standard definitions with local page evidence. This study investigates retrieval-conditioned language-model verification, language- and class-specific error patterns, and the retrieval limitations that arise when extending page-level checking to full reports. We evaluate a few-shot verifier on 793 non-Chinese test rows and separately analyze a reconstructed full-report retrieval diagnostic. The best text-prompt configuration reaches Macro F1 0.6140, compared with 0.4489 for an embedding baseline; a separate low-detail page-image configuration reaches 0.5748 under a setting in which demonstrations remain text-only. In the retrieval diagnostic, a BGE-M3 multi-representation first-stage arm improves Hit@1 from 0.220 to 0.266, while a separate GPT-5.4-nano reranking run reaches 0.328; these values are not additive. Comparative and qualitative analyses identify partial disclosure and language-dependent candidate recall as persistent boundaries. The academic contribution is a reproducible evidence-matching framework and a cross-system account of accuracy-complexity trade-offs; the practical implication is a staged workflow that exposes uncertainty and routes visually complex pages to targeted review.

Keywords

ESG compliance, SASB standards, retrieval-augmented generation, multilingual NLP, multimodal verification

Team Name

IMNTPU

Subtasks

Subtask 1 (multilingual evidence page retrieval) and Subtask 2 (page-level three-way compliance verification), covering five official languages: English, French, Japanese, Korean, and Thai.

1 Introduction

1.1 Background and Motivation

Sustainability reporting standards require firms to disclose environmental, social, and governance (ESG) information according to specific metric definitions, disclosure categories, and units of measure. The Sustainability Accounting Standards Board (SASB) standards are representative of this metric-oriented style: a disclosure is not only about a topic, but also about whether it satisfies a

precise metric, category, and unit requirement [17]. ESG reports are long, multilingual, and often semi-structured, so page-level compliance checking is costly for regulators, assurance providers, and analysts.

The NTCIR-19 RegCom task formulates this challenge as multinational, multilingual, and multi-industry regulatory compliance verification for ESG reports [9, 15]. In Subtask 2, each instance contains a SASB metric row and a single page from an ESG report. A system must predict three fields: `pred_label`, `category_match`, and `unit_match`. The main label has three values: `yes`, `yes but not complete`, and `no`. This makes the task harder than topical relevance classification. A page may discuss the same ESG topic yet fail to disclose the required quantity, omit sub-items, use a proxy, or report an incompatible unit.

Recent multilingual ESG and promise-verification studies demonstrate the value of evidence-aware analysis [3, 14, 16], while the current RegCom task emphasizes metric-level category and unit compatibility [15]. However, most reported comparisons still leave three questions open: whether retrieval-conditioned verification is statistically better than a transparent embedding baseline, how evidence modality affects difficult partial disclosures, and how retrieval errors propagate across languages.

1.2 Research Objective and Questions

This study aims to develop and critically evaluate a reproducible evidence-matching workflow for multilingual SASB compliance checking. Specifically, we ask:

- **RQ1:** Does the complete retrieval-conditioned few-shot system improve balanced classification performance over a multilingual embedding baseline on the same page-level rows?
- **RQ2:** How do model family, page-image evidence, disclosure class, and language change the observed performance and error patterns?
- **RQ3:** Which lexical, dense, multi-vector, and VLM retrieval components improve full-report page localization, and where does candidate recall remain the limiting factor?

1.3 Contributions

We present RegComEvidenceMatch, a lightweight evidence-matching verifier that uses labeled training rows as a retrieval memory and asks an LLM to make a structured page-level judgment. The design follows the intuition of few-shot prompting [2] and retrieval-augmented generation [7]: rather than updating model parameters, the system supplies task-relevant examples at inference time. Our contributions are fourfold. First, we provide a transparent retrieval-and-prompting design for SASB page-level verification. Second, we compare LLM and embedding-based

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systems under a common 793-row non-Chinese split and add paired confidence intervals and significance tests. Third, we combine quantitative ablations with language-level and case-based error analysis, including a reconstructed full-report retrieval extension. Fourth, we interpret the findings across the Cierpa and CSCU submissions and derive academic, practical, and managerial implications without treating non-aligned scores as a leaderboard. Because zero-shot and random-example controls were not run in this study, the results establish system-level performance differences rather than a causal gain attributable to retrieval alone.

1.4 Organization

Section 2 reviews ESG compliance, multilingual verification, retrieval augmentation, and related RegCom submissions. Section 3 describes the data, verifier, multimodal extension, and statistical validation. Section 4 reports overall, class-level, language-level, retrieval, and qualitative results. Section 5 discusses cross-system implications and limitations, and Section 6 concludes with academic contributions, managerial implications, and future research directions.

Beyond Subtask 2, Section 4.7 reports a preliminary retrieval extension toward the official Subtask 1 (evidence-page retrieval) setting, built from already-cached local runs. The extension is included to make the research process reproducible and to identify the remaining requirements for an end-to-end official evaluation.

2 Related Work

2.1 ESG Disclosure Analysis and Multilingual Verification

ESG disclosure analysis focuses on evaluating the completeness and reliability of environmental, social, and governance information in corporate sustainability reports [14]. Since these reports may be written in different languages and vary across regions, industries, and reporting formats, multilingual verification has become an important issue in sustainability analysis [8]. Researchers need to determine whether the disclosed content satisfies specific reporting standards and whether the meaning remains consistent across languages. This study focuses on the relationship between ESG reports and SASB metrics, using model-based methods to classify disclosures as complete, partially complete, or not disclosed.

Prior ESG NLP research has shown that domain-specific language models and evidence extraction are useful for sustainability and financial text. ClimateBERT adapts transformer language modeling to climate-related corporate and news text [19], while Kannan and Seki extract textual evidence for ESG scores from financial reports [6]. These studies motivate the use of report evidence rather than only document-level labels, but they do not directly address SASB metric-level compliance decisions.

2.2 Multilingual Promise and Compliance Verification

Multilingual promise and compliance verification has become an important research direction in ESG text analysis. Corporate sustainability reports are often published in different languages, and the same commitment or disclosure may vary across languages,

industries, and reporting formats. Therefore, models need to determine whether the commitment, supporting evidence, and standard requirements are semantically consistent. Recent work such as ML-Promise introduces multilingual corporate promise verification for ESG reports and studies evidence assessment across languages [16]. SemEval-2025 Task 6 further frames corporate promise verification as a multilingual and multi-industry task involving commitment detection and evidence consistency [3]. In contrast, NTCIR-19 RegCom focuses more on compliance verification by asking whether a single report page satisfies a specific SASB metric, including category and unit compatibility [9]. Such tasks therefore emphasize not only whether a topic is mentioned, but also whether the disclosure actually meets the required standard.

2.3 Retrieval-Augmented and Embedding-Based Methods

Our method is also related to few-shot prompting, retrieval-augmented generation, classical term weighting, and sentence embeddings. Large language models can perform new tasks from a small number of demonstrations [2], and retrieval-augmented generation supplies task-relevant external evidence at inference time [7]. We use TF-IDF because it is a strong, transparent lexical retrieval baseline for term-weighted document matching [13]. Sentence-BERT and multilingual knowledge distillation provide efficient embedding baselines for cross-lingual semantic matching [11, 12]. RegComEvidenceMatch combines these ideas by using retrieval to select few-shot cases for a structured single-page verifier.

2.4 Prompt-based Large Language Models for ESG Classification

Prompt-based Large Language Model classification uses natural language instructions to guide models in performing specific classification tasks without necessarily retraining model parameters. In ESG disclosure classification, disclosure standards, report segments, supporting evidence, and label definitions can be integrated into the prompt, allowing the model to judge whether a company’s disclosure satisfies a specific metric. Compared with traditional classifiers, prompt-based methods are more suitable for long, unstructured reports and diverse semantic expressions. When combined with a few examples or retrieved similar cases, the model can better understand the differences among labels such as complete disclosure, partial disclosure, and no disclosure. Therefore, prompt-based LLM classification provides a flexible approach for automated ESG report analysis.

2.5 Comparison with Other RegCom Submissions

Two other submitted systems provide useful contrasts to our page-text RAG design. Cierpa decomposes full-report compliance checking into relevant-page retrieval and label inference [4]. Its retrieval pipeline combines BM25, text embeddings, VLM-generated descriptions, image embeddings, reciprocal-rank fusion, and VLM reranking. For Subtask 2, it compares zero-shot inference with few-shot

and extended few-shot prompts, using either random or metric-prioritized example selection. This design tests both evidence localization and demonstration choice, but requires several embedding and generative components and incurs additional reranking cost.

CSCU instead emphasizes traceable evidence grounding [1]. Its Subtask 1 system uses table-of-contents-first section localization with a full-document fallback and a VLM-guided candidate verifier. For Subtask 2, it directly compares text-only, image-only, and text-plus-image page evidence. This design isolates the value of page modality more directly than our experiments, although its strongest retrieval variant jointly changes section selection and final page verification, so the gain cannot be attributed to a single component. In contrast, RegComEvidenceMatch assumes the task-provided page and concentrates on lightweight case retrieval and metric-level compliance reasoning from extracted text.

A terminology note is necessary before comparing these systems further: “Task 1” and “Subtask 1” are not used consistently across the three reports. We follow the official convention, in which RegCom Task 1 contains Subtask 1 (evidence-page retrieval within a full report) and Subtask 2 (single-page verification, the setting of this paper). CSCU’s usage matches this convention directly. Cierpa instead labels its full retrieval-plus-verification pipeline “Subtask 1” and treats page retrieval as a sub-problem within it, so a Cierpa “Subtask 1” number is not the same quantity as a CSCU or official Subtask 1 number. Section 4.7 of this report uses the official, CSCU-aligned sense of the term.

The two submissions provide two observations that directly qualify our interpretation. Cierpa finds that lexical, semantic, and visual retrieval signals are complementary, but that learned fusion weights can fail to transfer from training to test; its audit also shows that page-metric labels can be non-unique. CSCU reports a seen-document/unseen-query split and finds complementary value in text and page images, while cautioning that its VLM-guided retrieval changes several stages jointly. We use these findings as methodological context rather than as a paired cross-team ranking. Our paired Task 2 tests are reported in Table 3, and our retrieval-extension tests are reported in Section 4.7.

3 Methods

Subtask 2 is Single-Page Metric Verification. Given a single SASB metric and a single page from an ESG report, the objective is to determine whether that page contains relevant information and, if so, whether the information complies with the specified SASB category and unit of measure. Each instance contains structured SASB metadata, extracted page text, and three prediction targets: `pred_label`, `category_match`, and `unit_match`. The main label is yes, yes but not complete, or no; if the prediction is no, the auxiliary fields are normalized to N/A. Because only one page is available, the task requires fine-grained local evidence matching rather than full-report compliance analysis.

The experiments use the non-Chinese subset of the task package, covering English, French, Japanese, Korean, and Thai. The retrieval and classifier training pool contains 753 labeled rows, and the evaluation set contains 793 rows. The training pool contains 295 yes, 127 yes but not complete, and 331 no rows, while the test set contains 315 yes, 138 yes but not complete, and 340 no rows.

3.1 Proposed Research Architecture

Figures 1 and 2 show the two task-specific research architectures. Subtask 1 starts from full ESG reports and localizes evidence pages through multi-model retrieval and VLM reranking. Subtask 2 receives one metric and one page, retrieves labeled demonstrations, and performs three-way compliance verification with text and optional page-image evidence.

3.2 Retrieval-Augmented Few-Shot Verifier

RegComEvidenceMatch does not fine-tune the LLM. Instead, it treats the training CSV as a case library. For each test row, we build a retrieval document from the SASB topic, metric description, metric code, category, unit, key terms, `sasb_what_counts`, and the first 1,500 characters of extracted page text. A TF-IDF vectorizer with unigram and bigram features computes cosine similarity between the test row and all training rows [13].

The example selector retrieves three demonstrations. It first prioritizes rows with the same SASB metric code. Within that pool, it attempts to select the most similar example for each label: yes, yes but not complete, and no. If a label is missing for the same metric, the selector falls back to the global training pool. Remaining slots are filled by the most similar same-metric rows and then by the most similar global rows. This strategy is both similarity-based and label-balanced, reducing the chance that the prompt is dominated by the majority class.

The verifier prompt contains a compact system instruction, the retrieved demonstrations with gold outputs, the metric-specific label prior, the target SASB metadata, and the extracted PDF page text. The LLM is instructed to judge only the given page, not the full ESG report, and to return compact JSON containing `pred_label`, `category_match`, `unit_match`, short evidence, and a short reason. The prompt also states the boundary rules: relevant and complete disclosure is yes; partial, proxy-only, incomplete, or wrong-unit disclosure is yes but not complete; and semantically absent metric evidence is no.

3.3 Prompt Design

The implementation uses the same prompt structure for Gemini 2.5 Pro and GLM-5.2. The prompt is designed to make the model judge only the given page, calibrate its decision with retrieved examples, and return a valid task-format prediction. The following prompt template summarizes the main structure used in our system:

You are a SASB page-level verification assistant. Judge only the single PDF page provided. Return compact JSON only with the following fields:

```
{ "pred_label": "yes|yes but not complete|no",
  "category_match": "yes|no|N/A",
  "unit_match": "yes|no|N/A",
  "evidence": "short page evidence",
  "reason": "short reason" }
```

Rules:

- 1. Predict yes if the page contains relevant disclosure for the requested SASB metric.*
- 2. Predict yes but not complete if the page is relevant but partial, proxy-only, incomplete, or wrong-unit.*
- 3. Predict no if the exact metric is not semantically disclosed.*

Table 1: RegCom System Design Comparison: Task Focus, Page Evidence, Full-Report Page Retrieval, Retrieved Demonstrations, and Main Trade-Offs Across Three Submissions.

System	Task focus	Page evidence	Page retrieval	RAG examples	Main trade-off
RegComEvidenceMatch	Subtasks 1–2	Extracted text and page image	Lexical, dense, multi-vector, VLM reranking	TF-IDF, metric-first, label-balanced	Modular; end-to-end verifier coupling remains open
Cierpa	Subtasks 1–2	Text and page image	Lexical, semantic, visual, VLM reranking	Random or metric-prioritized	Broad signals; higher cost
CSCU	Subtasks 1–2	Text and/or page image	ToC-first, fallback, VLM guidance	Prompt-based verification	Traceable and multimodal; coupled changes

Subtask 1: Full-Report Evidence-Page Retrieval

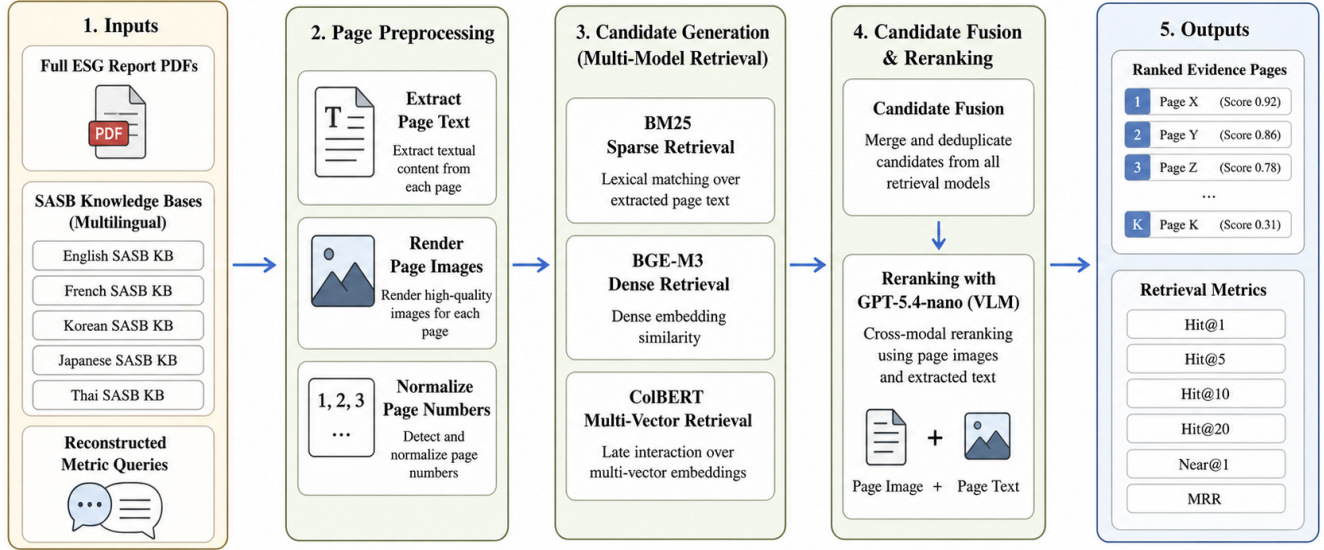


Figure 1: Subtask 1 Multilingual Full-Report Evidence-Page Retrieval Architecture.

4. If `pred_label` is no, set `category_match` and `unit_match` to N/A.
5. Use the retrieved few-shot examples and metric label distribution only as calibration. The target page evidence decides.
Retrieved few-shot examples:
Example 1–3: task metadata, evidence snippet, similarity score, same-metric flag, and gold output.
Now predict this test row:
Task: language, company/report id, file stem, page, metric code, topic, metric description, SASB category, unit, key terms, and what-counts guidance.
PDF page text: [extracted page text, truncated to 5,000 characters]
Return JSON only.

In the actual implementation, each retrieved example includes a page evidence snippet of up to 1,200 characters, its TF-IDF similarity score, whether it shares the same metric code as the target row, and the gold output. The prompt also includes the training label distribution for the same metric as a calibration prior. After generation, invalid or missing auxiliary labels are normalized by deterministic post-processing. For discussion-and-analysis metrics, relevant

narrative evidence is treated as category- and unit-compatible, and zero-event or no-incident statements can satisfy event-count metrics when the metric asks for occurrences or counts.

3.4 Compared Systems

We evaluate two LLM configurations using the same retrieval and prompt design. The first uses Gemini 2.5 Pro because the Gemini 2.5 model family is designed for advanced reasoning, multimodality, and long-context processing [5]. These capabilities are relevant to fine-grained page-level verification, where the model must compare local report evidence with a structured metric definition. The second uses GLM-5.2 through the GLM API as an alternative commercial LLM from the GLM family. Public GLM-family reports emphasize strong instruction following, long-context ability, and multilingual pretraining [20]. Both systems receive the same structured task fields, PDF page text, and three few-shot examples.

As a multimodal follow-up, we also run GPT-5.4-mini through the OpenAI Responses API. The target PDF page is rendered at 96 DPI and supplied as a low-detail image together with the extracted page text; the retrieved demonstrations remain text-only. This run

Subtask 2: Single-Page Metric Compliance Verification

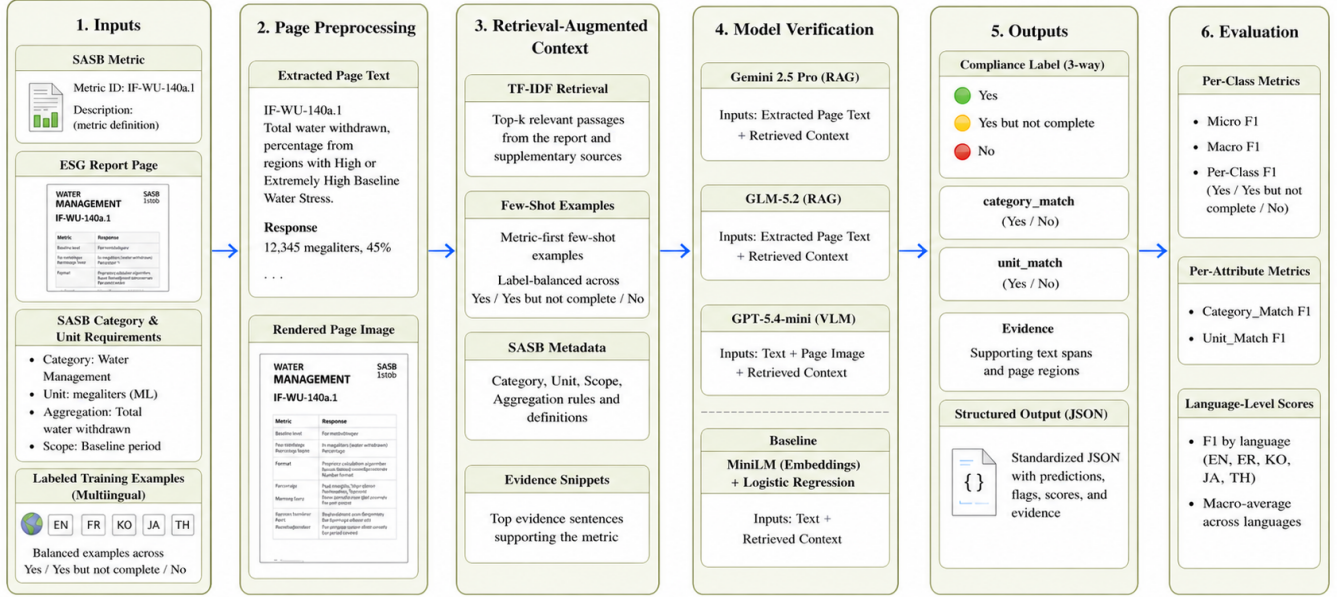


Figure 2: Subtask 2 Multilingual Single-Page Metric Compliance Verification Architecture.

is evaluated on the same 793 non-Chinese rows and is reported as a multimodal comparison, not as a replacement for the primary Gemini/GLM experiments.

As a lightweight open-source baseline, we train a MiniLM embedding classifier. MiniLM is suitable for this baseline because it compresses Transformer representations through self-attention distillation and is designed for efficient inference [18]. We encode the same SASB-page text representation with paraphrase-multilingual-MiniLM-L12-v2, a multilingual sentence-embedding model related to Sentence-BERT and multilingual knowledge distillation [11, 12]. A logistic regression classifier with balanced class weights predicts the three `pred_label` classes; we use it as a transparent linear classification baseline implemented with scikit-learn [10]. Auxiliary `category_match` and `unit_match` fields are derived deterministically from the predicted label for scoring-format compatibility. This baseline is fast and reproducible, but it cannot explicitly reason about completeness, units, or page-level disclosure boundaries.

3.5 Evaluation Metrics

We use Micro F1 and Macro F1 as the main evaluation metrics. Micro F1 aggregates true positives, false positives, and false negatives across all instances before computing the F1 score. It therefore reflects the overall classification performance on the evaluation set. Macro F1 computes the F1 score separately for each class and then averages the class scores. This metric is important for this task because yes but not complete is less frequent than yes and no, but it is central to compliance verification. Reporting both metrics allows us to evaluate both overall correctness and class-balanced

performance. For the retrieval extension, we additionally report exact page hits, `Near@1`, candidate recall, and MRR. Although the structured output also contains `category_match` and `unit_match`, the available exports do not provide independently validated categorical gold labels for these auxiliary fields. We therefore score `pred_label` and treat the auxiliary fields as format-normalized outputs rather than claiming separate category or unit F1.

3.6 Statistical Validation

For Task 2, prediction files are aligned by language, report identifier, topic, metric description, metric code, page, and file stem, with an occurrence index for exact duplicate rows. We compare every pair of systems on the same 793 rows using a two-sided exact McNemar test on correctness. We also compute paired nonparametric bootstrap confidence intervals (10,000 resamples, fixed seed) for accuracy and Macro F1 differences. The six pairwise McNemar p -values are adjusted with the Holm step-down procedure at $\alpha = 0.05$. For the reconstructed Task 1 retrieval experiments, we retain the paired bootstrap and McNemar analyses over the same 369 non-empty-gold query groups. These tests quantify uncertainty without treating independent aggregate scores as if they were paired observations.

3.7 Qualitative and User-Oriented Evaluation

We complement aggregate scores with case-based analysis of cached prediction traces. Cases are selected to represent baseline-to-model improvements, regressions, candidate-recall misses, index-page

Table 2: Task 2 Single-Page Metric Verification: Overall Micro F1 and Macro F1 for Four Models on 793 Non-Chinese Test Rows.

System	Micro F1	Macro F1
Gemini 2.5 Pro RAG	0.6507	0.6140
GLM-5.2 RAG	0.6280	0.5963
GPT-5.4-mini VLM	0.6217	0.5748
MiniLM + LR	0.4603	0.4489

confusions, and language-specific failures; the analysis reports the evidence pattern and the likely error mechanism rather than only the label. The released 65-row stratified sample is reserved for future double annotation and inter-annotator agreement. No human user study is claimed in the present report. A follow-up quantitative user evaluation is specified as a controlled comparison of compliance analysts using text-only versus multimodal evidence, measuring decision accuracy, time per row, abstention/calibration, and perceived trust on identical cases.

4 Experiments

4.1 Overall Results

Table 2 reports the overall results. Gemini 2.5 Pro achieves the best score, with 0.6507 Micro F1 and 0.6140 Macro F1. GLM-5.2 is close, with 0.6280 Micro F1 and 0.5963 Macro F1. GPT-5.4-mini with a low-detail page image obtains 0.6217 Micro F1 and 0.5748 Macro F1, below the two text-prompt LLM configurations in this run but well above MiniLM. MiniLM with logistic regression is much faster but substantially weaker, with 0.4603 Micro F1 and 0.4489 Macro F1. These results indicate that retrieval-conditioned LLM verification adds meaningful value beyond multilingual semantic embeddings, while a single low-detail image does not automatically improve the decision boundary for partial disclosures.

The paired tests qualify the apparent ranking in Table 2. After Holm correction across the six comparisons, each LLM configuration is significantly better than MiniLM on correctness, whereas the Gemini-GLM and Gemini-GPT-5.4-mini differences are not statistically significant at $\alpha = 0.05$. Thus, the point estimate favors Gemini, but the available paired evidence supports a robust LLM-versus-embedding contrast rather than a conclusive winner among the LLMs. The full row-alignment and resampling record is provided in artifacts/metrics/task2_paired_significance.json.

4.2 Class-Level Results

Table 4 shows class-level F1. The no class is strongest for both LLM systems, while yes but not complete remains the weakest class. Gemini reaches 0.4835 F1 on yes but not complete; GLM-5.2 reaches 0.4717; MiniLM reaches 0.3931. GLM-5.2 has higher recall on the partial class, but lower precision, suggesting that it more often uses the partial label as a cautious middle state. Gemini is more conservative on partial predictions and obtains better overall balance.

4.3 Language-Level Results

Language-level results are shown in Table 5. Gemini is strongest on French, Japanese, and Korean. Gemini and GLM-5.2 tie on Thai Micro F1, while GLM-5.2 has the highest Thai Macro F1. GPT-5.4-mini is strongest on English (Micro F1 0.6269 and Macro F1 0.5802), where Gemini is comparatively weakest; GPT-5.4-mini remains below Gemini on the other four languages. MiniLM is consistently below the LLM systems, with the largest gap on Thai.

4.4 Cross-Submission Comparison

Table 6 contextualizes our strongest result with the independently reported Subtask 2 results from Cierpa and CSCU. Cierpa reports an overall Macro F1 of 0.637 for its best metric-aware few-shot configuration across six languages [4]. CSCU reports 0.624 Macro F1 and 67.9% accuracy for text-plus-image verification over 967 retained page records [1]. Our Gemini configuration reports 0.614 Macro F1 and 65.07% Micro F1 over 793 non-Chinese rows; the additional GPT-5.4-mini VLM run reports 0.5748 Macro F1 and 62.17% accuracy on the same rows.

These values are close in magnitude but are not directly comparable. Our evaluation excludes Chinese, Cierpa applies a conflict-oriented cleaning procedure, and CSCU reports a retained 967-record set and excludes a documented source-alignment anomaly. The systems also differ in model, prompt, page modality, and example selection. Consequently, Table 6 is a comparison of reported experimental contexts rather than an official ranking or a paired significance test.

4.5 Result Analysis

The main finding is that the task benefits from explicit LLM-based verification. MiniLM embeddings capture broad multilingual similarity, but the task requires more than semantic closeness. The model must decide whether a page satisfies a specific SASB metric, whether the disclosure is complete, and whether the unit is compatible. This explains why the embedding classifier often fails despite using the same training rows and PDF text.

The comparison between Gemini 2.5 Pro and GLM-5.2 also illustrates that prompt-calibrated retrieval does not remove model-specific decision tendencies. GLM-5.2 predicts yes but not complete more frequently, which improves recall for partial disclosures but reduces precision. Gemini 2.5 Pro predicts no more often and obtains stronger overall F1 balance. GPT-5.4-mini with a low-detail page image is substantially stronger than MiniLM overall, but its partial-disclosure F1 (0.3686) remains below both text-prompt LLMs. This comparison is confounded: only the target page image was added, the demonstrations remained text-only, and image resolution/detail were fixed at 96 DPI/low detail. It therefore measures one multimodal configuration rather than the isolated causal contribution of page images. In practical compliance screening, the preferred model may depend on whether the user values recall for partial disclosures or conservative final acceptance.

The other submissions refine this interpretation. Cierpa reports that metric-prioritized few-shot examples improve its aggregate score over zero-shot inference, but the effect varies by language, with gains in some slices and reversals in others. Thus, examples can calibrate a difficult boundary but can also anchor the verifier

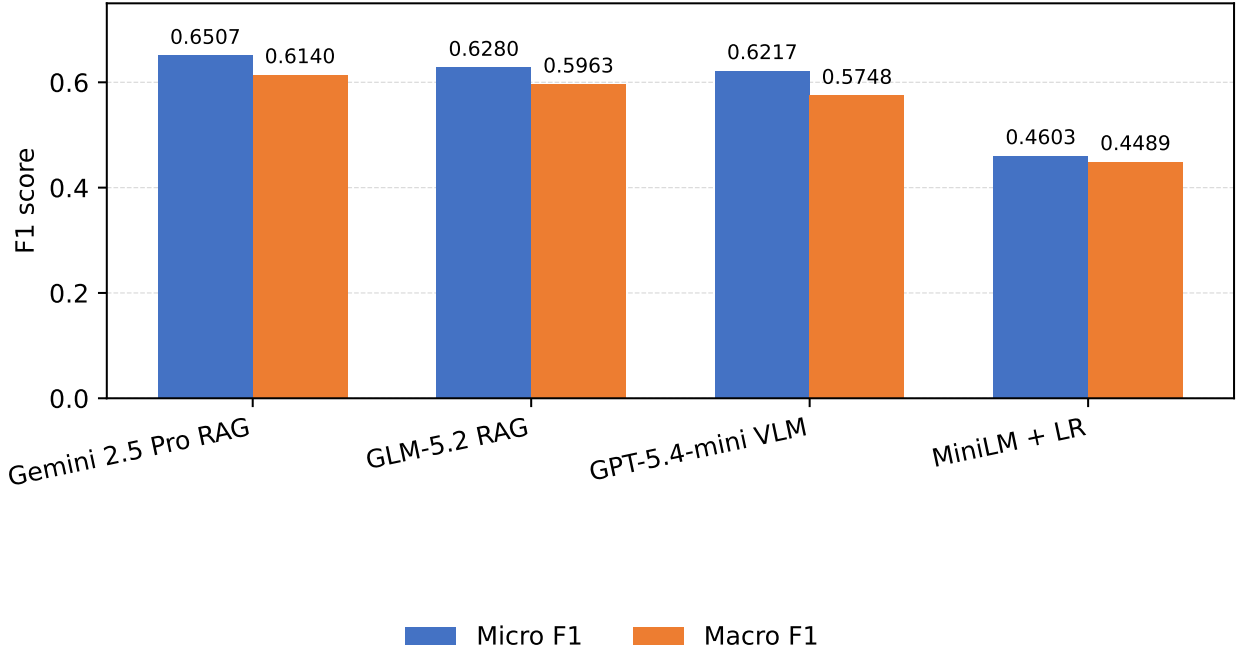


Figure 3: Task 2 Single-Page Metric Verification: Micro F1 and Macro F1 by Model on the 793 Non-Chinese Test Rows.

Table 3: Paired Task 2 Model Comparisons on Identical Non-Chinese Rows: Accuracy and Macro F1 Differences with 95% Paired Bootstrap Confidence Intervals and Holm-Adjusted Exact McNemar p -Values for Accuracy.

Comparison	Δ Accuracy		95% CI	Δ Macro F1		95% CI	Holm p (accuracy)
Gemini-GLM	+0.0227		[-0.0025, 0.0492]	+0.0177		[-0.0137, 0.0502]	0.209
Gemini-GPT-5.4-mini	+0.0293		[-0.0025, 0.0618]	+0.0395		[-0.0000, 0.0795]	0.294
Gemini-MiniLM	+0.1907		[0.1488, 0.2358]	+0.1653		[0.1204, 0.2111]	< 0.001***
GLM-GPT-5.4-mini	+0.0066		[-0.0240, 0.0366]	+0.0218		[-0.0119, 0.0553]	0.748
GLM-MiniLM	+0.1680		[0.1261, 0.2119]	+0.1476		[0.1058, 0.1916]	< 0.001***
GPT-5.4-mini-MiniLM	+0.1614		[0.1185, 0.2055]	+0.1259		[0.0814, 0.1702]	< 0.001***

Note: * p < 0.05, ** p < 0.01, and *** p < 0.001; stars mark two-sided exact McNemar tests after Holm correction across the six pairwise comparisons.

Table 4: Task 2 Single-Page Metric Verification: F1 Scores for the yes, yes but not complete, and no Prediction Classes on the Non-Chinese Test Set.

System	yes	ybnc	no
Gemini 2.5 Pro RAG	0.6327	0.4835	0.7258
GLM-5.2 RAG	0.5919	0.4717	0.7250
GPT-5.4-mini VLM	0.6266	0.3686	0.7292
MiniLM + LR	0.5133	0.3925	0.4411

Note: ybnc = yes but not complete.

Table 5: Task 2 Single-Page Metric Verification: Language-Level Micro F1 and Macro F1 for English, French, Japanese, Korean, and Thai.

System	Metric	English	French	Japanese	Korean	Thai
Gemini 2.5 Pro RAG	Micro F1	0.5473	0.6771	0.7231	0.7718	0.5537
	Macro F1	0.4833	0.6382	0.6801	0.7467	0.5129
GLM-5.2 RAG	Micro F1	0.5473	0.6406	0.6538	0.7584	0.5537
	Macro F1	0.5090	0.5971	0.5996	0.7337	0.5180
GPT-5.4-mini VLM	Micro F1	0.6269	0.6354	0.6077	0.6711	0.5455
	Macro F1	0.5802	0.5825	0.5555	0.6321	0.4864
MiniLM + LR	Micro F1	0.4726	0.5104	0.4846	0.4430	0.3554
	Macro F1	0.4550	0.4548	0.4767	0.4388	0.3100

to inappropriate edge cases. This observation supports our use of

same-metric examples while motivating the missing zero-shot and random-example controls.

Table 6: Cross-Submission Context: Independently Reported Best Subtask 2 Results with Evaluation Scope and Retained-Row Differences.

System	Macro F1	Acc./Micro F1	Scope
Cierpa Best F1	0.637	–	Six languages, cleaned
CSCU text+image	0.624	0.679	967 retained rows
RegComEvidenceMatch Gemini	0.614	0.6507	793, no Chinese
RegComEvidenceMatch GPT-5.4-mini VLM	0.5748	0.6217	793, no Chinese

CSCU provides complementary evidence about input modality: its text-plus-image setting outperforms text-only, although image-only and text-plus-image intervals overlap. This suggests that visual evidence can recover structure lost during extraction, but it does not identify the isolated contribution of an image under our different prompt, resolution, and model settings. Demonstration relevance and evidence fidelity therefore address different failure sources: the former calibrates the decision boundary, whereas the latter determines whether the required value, unit, and scope are visible.

Cierpa’s end-to-end analysis further shows that retrieval and label inference can compound errors when page-metric labels are non-unique. A correctly retrieved page may still be scored against a mismatched page label when the same report contains both positive and negative instances for one metric. Any future end-to-end version of RegComEvidenceMatch should therefore preserve row-level identity and avoid attributing every chained error to retrieval alone.

4.6 Error Analysis

Errors remain concentrated around partial disclosure. A page may contain relevant ESG language but omit a required denominator, scope, disaggregation, event count, or unit. Conversely, a page may present a SASB index that names a metric but does not contain the underlying disclosure. Single-page evaluation makes these cases difficult because evidence may appear on neighboring pages. The current system also relies on extracted PDF text; tables or figures that are poorly extracted may be underused.

The observed errors can be summarized into three common patterns. First, topical matches are sometimes confused with compliant disclosures: the page discusses the correct ESG issue but does not provide the metric-specific quantity, category, or unit. Second, partial disclosures are hard to separate from complete disclosures when the page provides proxy evidence, percentages without denominators, or qualitative descriptions without the required numerical detail. Third, extracted text from PDF tables may lose row or column structure, making it difficult for the verifier to connect a number with the correct SASB metric.

These patterns recur across submissions. Cierpa reports that few-shot examples improve some language-level partial-disclosure results but reverse in others, indicating that the yes but not complete boundary is sensitive to demonstration choice. CSCU likewise identifies this label as the hardest class and attributes remaining errors to incomplete page evidence and confusion between full and partial disclosure. Its Subtask 1 analysis additionally shows that many retrieval errors fall on a page adjacent to the annotated page. Although our system receives the official target page and does not evaluate retrieval, this finding exposes a limitation of strict single-page verification: the verifier can be penalized

for missing evidence that is structurally associated with the target disclosure but printed on a neighboring page. A consolidated, cross-method error taxonomy and worked examples are provided as supplementary error-analysis material.

4.7 Task 1 Retrieval Extension

The RegCom shared task also defines Subtask 1 as full-report compliance matching: a system must localize relevant pages and then verify the disclosure category and unit. The experiments above do not evaluate that setting because Subtask 2 supplies the task-provided page directly. We therefore reconstruct report-metric query groups from the available page-level files and evaluate the page-localization stage separately.¹ Since the official Subtask 1 input records, gold page sets, empty/no-disclosure rules, and evaluator are not included in this checkout, the scores below should be interpreted as reconstructed retrieval measurements rather than official leaderboard values.

The retrieval pipeline follows the general shape of Cierpa’s retrieval design (Section 2.5): independent lexical and dense candidate generators, weighted reciprocal-rank fusion, a lane-union candidate pool, and Top-10 vision-language-model (VLM) reranking, evaluated against the reconstructed query groups. On 490 report-metric groups (369 with at least one non-empty gold page, 121 without), the frozen 10-candidate dense-fused configuration reaches Hit@1 0.220, Hit@5 0.493, Hit@10/candidate recall 0.631, and MRR 0.350 on the non-empty groups. A separate gpt-5.4-nano reranking run over those same 10 candidates reaches Hit@1 0.328, Hit@5 0.583, and MRR 0.431; these are not additive with the local BGE-M3 ablation reported below, and Hit@10 remains 0.631 because reranking cannot recover a page absent from the candidate pool. The dense-fused 10-candidate candidate recall@10 varies sharply by language (English 0.891, French 0.840, Korean 0.686, Chinese 0.576, Japanese 0.545, Thai 0.313), which identifies Thai, Chinese, and Japanese candidate generation, rather than reranking quality, as the current bottleneck. The 490 VLM traces completed successfully after rate-limit retries, using 6,758,052 input and 143,304 output tokens at an estimated cost of US\$1.531. A separate cached verifier trace produced a predicted-label distribution of 301 no, 143 yes but not complete, and 46 yes; only a small 74-row subset joined unambiguously to available truth after accounting for label non-uniqueness, so we do not treat that exploratory accuracy as a general Task 1 estimate.

A deeper-candidate follow-up configuration (task1-neighbor2: the same dense-enabled fusion, plus a two-page neighbor expansion around the Top-10 anchors, doubling the VLM candidate pool to Top-20) was completed on all 490 groups. The neighbor2 candidate-pool recall reaches 0.710 at Top-20 (0.789 at Top-30), while its 20-image nano reranker reaches Hit@1 0.314, Hit@5 0.561, Hit@10 0.648, Near@1 0.398, and MRR 0.423, compared with Hit@1/5/10 0.328/0.583/0.631 and MRR 0.431 for the 10-image primary configuration. The larger pool therefore improves coverage at Hit@10 but slightly reduces Top-1 and MRR. The effect is language-dependent: Chinese, Japanese, Korean, and Thai Top-1 improve in the 20-image

¹Full run configurations, dates, and the claim-boundary policy are maintained in docs/EXPERIMENT_LOG.md and docs/TASK1_VLM_PLAN.md in the project repository.

neighbor2 run (e.g. Japanese 0.182 to 0.309), but French Top-1 regresses from 0.320 to 0.300 (MRR 0.484 to 0.467) even though French candidate recall rises from 0.840 in the 10-image primary run to 0.900 in neighbor2. Consequently neither configuration strictly dominates the other. The complete neighbor2 run used 13,178,046 input and 275,396 output tokens at an estimated cost of US\$2.980.

The completed local BGE-M3 ablation uses the same reconstructed query groups and page mapping and runs without API calls. It compares dense retrieval, dense+sparse fusion, dense+sparse+ColBERT multi-vector fusion, and BGE-Reranker-v2-M3 reranking. Table 7 reports the unified scores on the 369 query groups with non-empty gold pages; the full reconstructed input contains 490 groups, including 121 empty-gold groups.

The best first-stage configuration is the three-representation BGE-M3 fusion, improving over the MiniLM baseline by 4.6, 5.2, and 4.4 percentage points on Hit@1, Hit@5, and Hit@10, respectively, and by 0.048 MRR. The gains are consistent with the component ablation: sparse features add about one point over dense-only retrieval, and the ColBERT-style multi-vector lane adds a further 1.5–2 points. Natural-language query rewrites underperform the original metric-keyword queries (0.209–0.228 Hit@1 across the tested variants), suggesting that the added English “what counts” wording introduces noise for non-English reports. Increasing the BGE maximum length from 2,048 to 8,192 tokens has almost no effect because most page texts are shorter than the limit. In contrast, the BGE-Reranker-v2-M3 arm lowers pooled Top-1 and MRR to 0.160 and 0.313: it often promotes SASB/GRI index pages that mention a metric code over the page containing the actual disclosure. Language-specific results show the largest Top-1 gains for Korean (0.216 to 0.373) and Japanese (0.182 to 0.236); in the BGE-M3 three-representation/raw arm, Thai Hit@10 is 0.281 and remains the principal candidate-generation bottleneck. These findings support stronger multilingual candidate generation, but also show that reranking quality and disclosure-page relevance must be analyzed separately. A 65-row stratified, human-review-only error sample is provided as supplementary material; its qualitative judgment columns intentionally remain blank for manual annotation.

To quantify uncertainty, we used 10,000 paired bootstrap resamples over the same 369 non-empty groups. The best BGE fusion improves Hit@1, Hit@5, Hit@20, and MRR significantly relative to the baseline, while the positive Hit@10 and Near@1 differences include zero in their 95% intervals. A paired McNemar test on Hit@1 gives 22 baseline-only versus 39 BGE-only successes ($p = 0.040$). The raw-versus-natural query comparison is significant for the multi-vector system (22 versus 8, $p = 0.016$), and the all-language reranker significantly degrades Top-1 (58 versus 25, $p = 0.0004$); language gating recovers a significant portion of those errors (14 versus 48, $p < 0.001$).

The error analysis separates candidate recall from ranking. The reranker selects a SASB/GRI index page as Top-1 for 91/369 raw queries (66 are incorrect) and 142/369 natural queries (111 incorrect). Representative failures include an esun query whose gold page 14 is replaced by a GRI index at page 195, and two hyundai metrics whose gold pages 72–73 are both replaced by the same SASB index at page 114. Thai and French fail for different reasons: the BGE-M3 three-representation/raw arm has Thai Hit@10 of 0.281, and the 20-image neighbor2 VLM pool misses the Thai gold page from Top-20

for 32/64 groups; by contrast, the 10-image primary VLM run has French candidate recall@10 of 0.840, neighbor2 raises it to 0.900, and the primary run’s French Hit@10 is 0.860, so the dominant French issue is ordering rather than recall (29/50 gold pages are in Top-20 but not Top-1). The 121 empty-gold groups also expose a limitation of the current interface: the retriever always returns candidates, so abstention detects 0/121 cases, and the fused Top-1 score has AUC 0.48 for distinguishing empty from non-empty groups. Finally, three out-of-range gold pages in the SPT report are caused by a one-page extraction failure; because every arm uses the same page map, they affect comparisons equally.

5 Discussion

5.1 Strengths and Weaknesses across Approaches

Table 8 summarizes the strengths and weaknesses each submission reports for itself. The three approaches occupy different points in the accuracy–complexity space. RegComEvidenceMatch now has a modular text-first path plus optional BGE-M3 retrieval and GPT-5.4-mini page-image verification. This keeps the core pipeline easy to inspect and comparatively inexpensive, while allowing visual evidence for pages whose layout is poorly represented by extracted text. Its central weakness is that the optional multimodal path still uses a single low-detail image and does not fully preserve cross-page context.

Cierpa offers the broadest retrieval design: lexical, semantic, and visual rankings capture complementary signals, but the larger pipeline requires multiple representations and model calls. Its few-shot study is particularly informative for our setting because metric alignment is useful overall but not uniformly beneficial across languages.

CSCU contributes a strongly traceable design and an explicit modality comparison. Its results suggest that whole-page visual evidence can recover information lost by text extraction, while its retrieval analysis distinguishes candidate-generation failures from verifier failures. Its VLM-guided variant changes ToC checking, section selection, and final verification together, so the improvement measures a package of interventions rather than an isolated ablation. These observations motivate our staged, explicitly confounded multimodal comparison.

5.2 Evaluation Scope and Validity

Macro F1 is more informative than accuracy alone because the partial class is less frequent and consistently difficult. Nevertheless, a single aggregate score hides material differences in language coverage, class distribution, evidence quality, and retained records. Cross-submission scores should therefore be accompanied by their evaluation scope, as in Table 6, and should not be compared as if generated on an identical paired sample.

The released reference data also complicate evaluation. Cierpa’s own audit of the official release quantifies this: across all six languages, 21.1% of training and 16.5% of test relevant-page targets are non-unique for the same metric tuple, and 55.6% of training and 57.2% of test instances share a non-unique gold label at the PDF level, because a no label marks absence from one page rather than from the whole report. An evaluator that joins rows only by a

Table 7: Task 1 Reconstructed Full-Report Page Retrieval: Ablation Results for Five Retrieval Configurations on 369 Non-Empty-Gold Query Groups.

Configuration	Hit@1	Hit@5	Hit@10	Hit@20	Near@1	MRR
MiniLM fused baseline	0.220	0.493	0.631	0.751	0.293	0.350
BGE-M3 dense	0.238	0.504	0.631	0.748	0.290	0.369
BGE-M3 dense+sparse	0.249	0.523	0.648	0.772	0.312	0.381
BGE-M3 dense+sparse+ColBERT	0.266	0.545	0.675	0.818	0.325	0.398
BGE-Reranker-v2-M3 (raw, all languages)	0.160	0.504	0.672	–	–	0.313

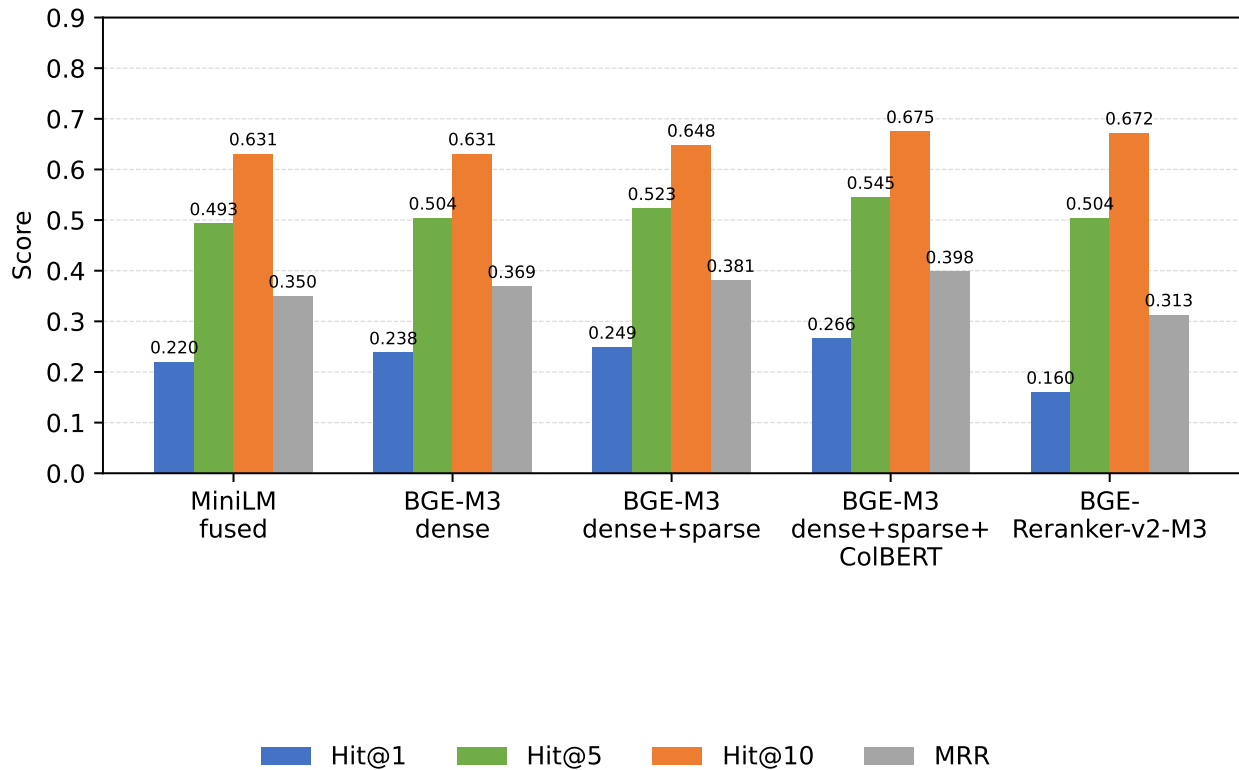


Figure 4: Task 1 Reconstructed Full-Report Page Retrieval: Hit@1, Hit@5, Hit@10, and MRR by Retrieval Configuration on 369 Non-Empty-Gold Query Groups.

non-unique tuple can collapse instances and change the reported result. Reviewer-facing analysis should preserve row-level identity or introduce a unique sample identifier. Separately, CSCU’s audit of the official Subtask 1 split found that all official test documents also appear in the training split, which it reports as seen-document, unseen-query evaluation rather than unseen-document generalization; because Cierpa, CSCU, and RegComEvidenceMatch all draw on the same official RegCom release, this framing plausibly applies to the other submissions’ splits as well, though we have not independently confirmed it for the non-Chinese CSV split used here. Moreover, the available reference files do not provide independently validated gold labels for the predicted `category_match` and `unit_match` fields; these auxiliary outputs should not be scored by

treating an answer-unit string as an equivalent categorical gold label.

Finally, exact single-page evaluation is intentionally controlled but does not represent every real report. Tables may continue across pages, narrative definitions may precede a numerical table, and SASB index pages may point to rather than contain evidence. Future evaluation could complement exact-page labels with acceptable page sets, neighboring-page sensitivity, or short evidence-span annotations. These alternatives should supplement rather than silently replace the official protocol.

Table 8: Cross-System Comparison: Self-Reported Strengths and Weaknesses of the Three RegCom Submissions.

System	Strengths	Weaknesses
RegCom-Evidence-Match	Transparent and label-balanced examples; optional GPT-5.4-mini page-image verifier; reproducible and modular	Single-page image is low-detail and loses cross-page context; multimodal run is below Gemini on Macro F1; no zero-shot or random-example control; non-Chinese scope only
Cierpa	Multi-signal retrieval (lexical, semantic, visual) with RRF; explicit dataset non-uniqueness audit; systematic few-shot ablation (ZI/FI/xFI × random/metric-prioritized)	Trained fusion weights underperform untuned fusion on test; VLM reranking has higher cost; few-shot gains are language-dependent and can reverse
CSCU	Single-factor modality ablation (text/image/text+image); bootstrap CIs and paired significance tests; explicit seen-document, unseen-query framing	VLM-Guided variant changes section selection and final verification jointly, so the gain is not attributable to one component; Subtask 1 and 2 evaluated independently, not end-to-end; exact-page scoring penalizes adjacent-page evidence

5.3 Implications for System Design

The comparison suggests a modular research direction. Metric-prioritized examples can improve decision calibration, while page images can improve evidence fidelity; neither substitutes for the other. A practical extension of RegComEvidenceMatch would preserve its lightweight text path for narrative pages and route table-heavy, scanned, or low-text pages to a multimodal verifier. A controlled ablation among zero-shot, random few-shot, TF-IDF few-shot, and metric-prioritized few-shot settings would then isolate the contribution of retrieval separately from the underlying LLM. Such experiments are preferable to attributing all gains to “RAG” when the model, prompt, sample scope, and evidence modality also differ.

5.4 Limitations

Our conclusions are subject to several limitations. First, our primary results cover 793 non-Chinese rows and therefore do not constitute a complete six-language evaluation. Second, Gemini 2.5 Pro and GLM-5.2 are remote models whose behavior and availability may change, while exact generation-level seeds and provider-side versions were not preserved. Third, our current comparison includes two LLMs and one embedding classifier but does not isolate the effect of retrieved demonstrations against zero-shot and random-example controls; consequently, the observed LLM-versus-baseline gap should not be interpreted as a causal RAG gain. Fourth, page evidence is extracted as text, so tables, figures, reading order, and

scanned content may be degraded. Fifth, the GPT-5.4-mini comparison uses one low-detail target-page image while demonstrations remain text-only, so it is a multimodal configuration study rather than an isolated modality ablation. Sixth, raw scores from the other submissions are contextual evidence rather than paired comparisons because their cleaning, retained samples, and modalities differ; the paired tests in Table 3 apply only to our aligned 793-row exports and cannot establish an official cross-team ranking. Seventh, the preliminary Task 1 retrieval extension in Section 4.7 is diagnostic only: the official evaluator and gold page-set contract are not yet confirmed, so none of those numbers should be read as official. Finally, our qualitative error taxonomy identifies plausible mechanisms but does not yet provide a fully annotated error sample with inter-annotator agreement or a completed human user study.

6 Conclusions

6.1 Key Findings

This study addressed three research questions. For RQ1, retrieval-conditioned LLM verification substantially outperformed the MiniLM baseline on the common 793-row split, and the paired tests confirmed the LLM-versus-embedding contrast after Holm correction; differences among the LLMs themselves were not statistically conclusive. For RQ2, the yes but not complete class and Thai-language rows remained the main weaknesses, while a single low-detail page image improved evidence access but did not match the strongest text-prompt result. For RQ3, BGE-M3 dense, sparse, and multi-vector signals improved reconstructed page retrieval, whereas a generic reranker frequently promoted index pages and could not recover candidate-generation misses.

6.2 Academic Contributions

The primary academic contribution is an evidence-matching formulation that connects SASB metric definitions, page-level evidence, disclosure completeness, and category/unit compatibility in one reproducible workflow. A second contribution is methodological: the study combines controlled ablations, row-aligned paired tests, confidence intervals, language slices, and case-based error analysis instead of relying on a single aggregate score. A third contribution is comparative. By placing our results beside Cierpa and CSCU, we show that retrieval breadth, evidence fidelity, and prompt calibration are complementary design dimensions, and that partial disclosure is a shared theoretical boundary across systems.

6.3 Managerial and Practical Implications

The managerial implication is a risk-aware review pipeline for regulators, assurance providers, and sustainability teams. A lightweight text verifier can triage routine narrative pages, while low-text, scanned, or table-heavy pages can be routed to multimodal review. Retrieval scores and abstention signals should be exposed to analysts because a high topical similarity does not guarantee a compliant quantity or unit. The practical implication is that organizations should retain row-level evidence, model/version metadata, and uncertainty intervals when deploying automated compliance checks; this supports auditability, targeted human review, and more defensible management decisions.

6.4 Future Research

Future work should complete an official end-to-end Subtask 1 evaluation, add OCR and neighboring-page context, and isolate zero-shot, random-example, and metric-prioritized retrieval effects. A controlled qualitative study with double annotation can quantify error-category agreement, while a quantitative user evaluation should compare analysts using text-only and multimodal evidence on identical cases, measuring decision accuracy, review time, calibration/abstention, and perceived trust. These extensions would test whether the observed model improvements translate into reliable human-AI compliance decisions rather than only higher offline scores.

Resource Availability

The supplementary resources for this report, including the source code, public reproducibility inputs, evaluation scripts, and model outputs/predictions, are available at: <https://github.com/wesleyseawatch2/sasb-page-rag-regcom> and <https://drive.google.com/drive/folders/1z5DGv2l4TL9sw9BCBBohASZp3HHLj5Uo?usp=sharing>.

Acknowledgments

This research was supported by the National Science and Technology Council (NSTC), Taiwan, under Grants NSTC 114-2425-H-305-003, NSTC 114-2221-E-305-002, and NSTC 115-2221-E-305-012; the Industrial Technology Research Institute (ITRI) and National Taipei University (NTPU), Taiwan, under Grants NTPU-114A513E01 and NTPU-113A513E01; National Taipei University under Grant 114-NTPU_ORDA-F-004; and the Ministry of Education (MoE), Taiwan, through the AI CUP 2026 ESG Sustainability Commitment Verification Competition and Annotation Data Collection Project and the Physical AI and Physical AI Robotics Curriculum Development Project under the Forward-Looking AI Talent Cultivation Program.

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